Lab Review Session

Pseudo-code Warm-up:

Anonymized data from a 2017 study on the effect of the higher education system on upward mobility (Chetty et al. 2017).

```
load("data/colleges.Rdata")
colleges$name <- substr(colleges$name, 1, 22)
head(colleges[, c("region", "state", "tier_name", "name", "mr_kq5_pq1")])
```

```
  region state                      tier_name                   name  mr_kq5_pq1
1      3    TX            Selective private Abilene Christian Univ 0.014363836
2      3    GA  Nonselective 4-year public Abraham Baldwin Agricu 0.014857335
3      4    CO             Selective public Adams State University 0.018844681
4      1    NY            Selective private     Adelphi University 0.032585073
5      2    MI            Selective private         Adrian College 0.009564629
6      3    FL Nonselective 4-year private Adventist University O 0.037858434
```

- `region`: 1 for Northeast, 2 for Midwest, 3 for South, and 4 for West
- `state`: State name
- `tier_name`: Tier defined by selectivity and type
- `mr_kq5_pq1`: Mobility rate, top 20% of the income distribution.

Give pseudo-code to answer: what states in the west have the highest average mobility rate across their private and elite colleges?

Full points answer:

1. Use `filter()` to subset the `colleges` data set to rows where `tier_name` is either "Selective private" or "elite" **AND** where `region` = 4
2. Use `group_by(state)` to perform the following operation for each value of `state`
3. Use `summarize()` to calculate the average `mr_kq5_pq1` value per `state`
4. Use `arrange()` to sort states from highest to lowest average `mr_kq5_pq1`

- *Note that this answer:*
  - *Names specific `dplyr` functions*
  - *Briefly demonstrates understanding of what those functions do, in context*
  - *Refers to the specific variable names in the data set*

Pseudo-code example 1

Looking at `pdxtrees`.

```
# A tibble: 5 x 7
  Genus          DBH Condition Tree_Height Crown_Width_NS Park              Native
  <chr>        <dbl> <chr>           <dbl>          <dbl> <chr>             <chr>
1 Quercus        3.3 Fair               16             14 Kenilworth Park   No
2 Pseudotsuga   43.1 Fair              148             61 Kenilworth Park   Yes
3 Thuja         14.1 Fair               42             21 Kenilworth Park   Yes
4 Acer          36.5 Good               80             68 Woodstock Park    No
5 Pinus         10.3 Fair               27             24 Sellwood Riverf~ No
```

Write pseudo-code to: Make a contingency table of whether trees in Woodstock Park are "big guys" or "little guys" by whether they are native species or not. Trees are considered "big guys" when DBH > 30 or Tree_Height > 100 and "little guys" otherwise.

Example pseudo-code to write as your response:

1. Use `mutate()` and `case_when()` to create a new variable that is "big guy" for rows with `DBH` > 30 **or** `Tree_Height` > 100, and "little guy" otherwise.
2. Use `filter()` to only consider rows where `Park` == "Woodstock Park"
3. Use `count(guy_type, Native)` to make the final contingency table.

Actual code for reference, but not expected in your response

```
near_Reed %>%
  mutate(guy_type = case_when(DBH > 30 | Tree_Height > 100 ~ "big guy",
                              TRUE ~ "little guy")) %>%
  filter(Park == "Woodstock Park") %>%
  count(guy_type, Native)
```

```
# A tibble: 4 x 3
  guy_type    Native     n
  <chr>       <chr>  <int>
1 big guy     No        26
2 big guy     Yes       67
3 little guy  No        16
4 little guy  Yes       29
```

Pseudo-code example 2

```
  species    island bill_len bill_dep flipper_len body_mass    sex year
1  Adelie Torgersen     39.1     18.7         181      3750   male 2007
2  Adelie Torgersen     39.5     17.4         186      3800 female 2007
3  Adelie Torgersen     40.3     18.0         195      3250 female 2007
4  Adelie Torgersen       NA       NA          NA        NA   <NA> 2007
5  Adelie Torgersen     36.7     19.3         193      3450 female 2007
6  Adelie Torgersen     39.3     20.6         190      3650   male 2007
```

Write pseudo-code to: Remove rows where neither `body_mass` nor `sex` is missing, and investigate (by eye) whether there are differences in median `body_mass` or `bill_len` by `sex` within the same `species`. Order the result by species.

Example pseudo-code to write as your response:

1. Use `filter()` to remove rows of `penguins` where either `body_mass` or `sex` are NA.
2. Use `group_by(sex, species)` to do the following computation for male and female penguins within each `species`.
3. Use `summarize()` to calculate the median `body_mass` and median `bill_len` within each species-sex group.
4. Use `arrange(species)` to order the result by species.

Actual code for reference, but not expected in your response

```
penguins %>%
  filter(!is.na(body_mass), !is.na(sex)) %>%
  group_by(sex, species) %>%
  summarize(med_mass = median(body_mass),
            med_len = median(bill_len)) %>%
  arrange(species)
```

```
# A tibble: 6 x 4
# Groups:   sex [2]
  sex    species   med_mass med_len
  <fct>  <fct>        <dbl>   <dbl>
```

1	female	Adelie	3400	37
2	male	Adelie	4000	40.6
3	female	Chinstrap	3550	46.3
4	male	Chinstrap	3950	51.0
5	female	Gentoo	4700	45.5
6	male	Gentoo	5500	49.5

Explaining code example 1

```
biketown %>%
  mutate(type = case_when(Duration > 60 & PaymentPlan == "Subscriber" ~ "long subscriber",
                          Duration > 60 & PaymentPlan == "Casual" ~ "long casual",
                          TRUE ~ "other")) %>%
  filter(type != "other") %>%
  group_by(type) %>%
  summarize(mean_dist = mean(Distance_Miles))
```

Explain what each line of code does in context, and sketch what the final data frame would look like.

1. The lines with `mutate()` create a new variable `type` that is "long subscriber" for subscriber rides where the ride duration was over an hour, "long casual" for casual rides where the ride duration was over an hour, and "other" for all other rows.
2. The line with `filter()` removes rows where `type` is "other".
3. The last two lines calculate the average distance of rides for long subscriber and long casual rides.

```
# A tibble: 2 x 2
  type            mean_dist
  <chr>               <dbl>
1 long casual          5.75
2 long subscriber      4.90
```

Explaining code example 2

```
penguins %>%
  filter(!is.na(sex),
         island %in% c("Biscoe", "Dream")) %>%
  count(species, island, sex) %>%
  group_by(species, island) %>%
  mutate(prop = n / sum(n)) %>%
  arrange(species, island, sex)
```

Explain what each line of code does in context, and sketch what the final data frame would look like. Reminder that there are 3 different species in `penguins`: Adelies, Gentoos, and Chinstraps.

1. Keep rows where `sex` is not missing and where penguins are either on the Dream or Biscoe islands.
2. Get counts of how many penguins on each island broken down by sex and species.
3. For each species on each island, calculate the proportion of male and female penguins.
4. Sort the result by species, island, and sex.

```
# A tibble: 8 x 5
# Groups:   species, island [4]
  species   island sex        n  prop
  <fct>     <fct>  <fct>  <int> <dbl>
1 Adelie    Biscoe female    22 0.5
2 Adelie    Biscoe male      22 0.5
3 Adelie    Dream  female    27 0.491
4 Adelie    Dream  male      28 0.509
5 Chinstrap Dream  female    34 0.5
6 Chinstrap Dream  male      34 0.5
7 Gentoo    Biscoe female    58 0.487
8 Gentoo    Biscoe male      61 0.513
```

Ggplot code + deconstruction 1

Part A

Identify (i) what value(s) of `geom_---()` are involved in this graph and (ii) what you expect to be inside of `aes()`.

(i) This combines `geom_boxplot()` and `geom_jitter()` (saying `geom_point()` would be fine too since we haven't used jittering much).

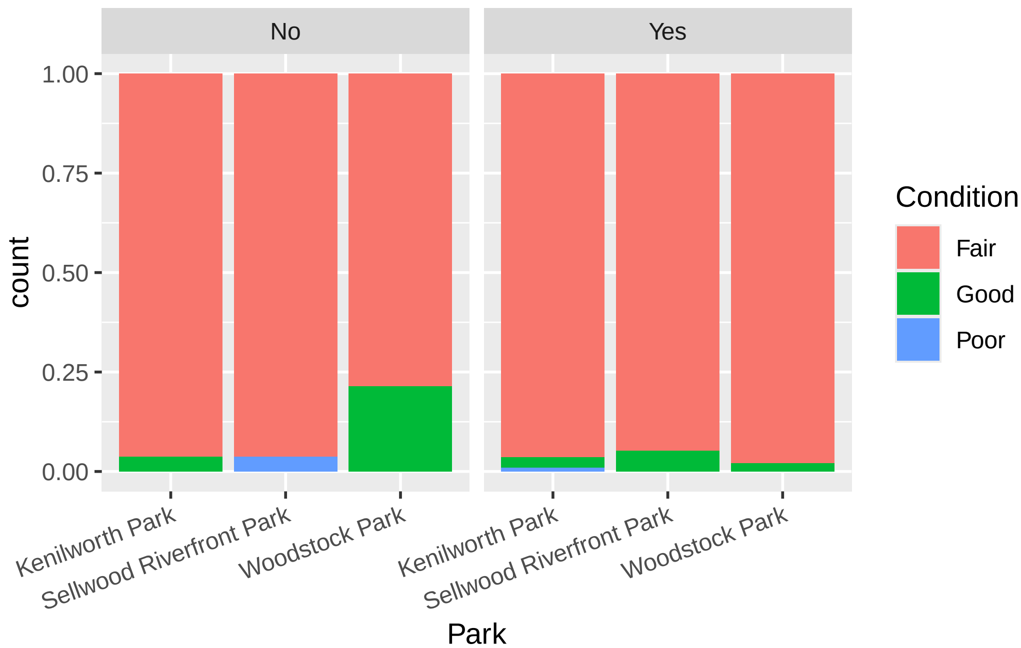
(ii) Aesthetic mappings include `x = Park`, `y = Tree_Height`, `color = Native`.

Part B

(i) What type of graph will the following code produce, and (ii) how will all variables are mapped to aesthetic features? It might help to sketch an example of what the graph could look like.

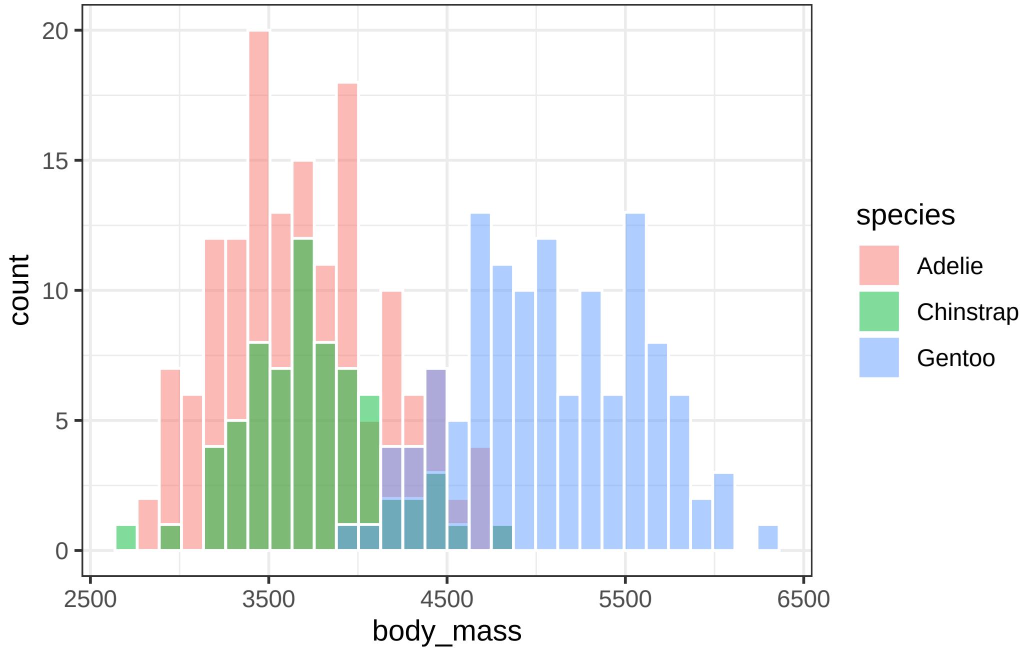
```
ggplot(near_Reed, aes(x = Park, fill = Condition)) +
  geom_bar(position = "fill") +
  facet_wrap(~Native)
```

(i) This produces a barplot. More specifically, a "filled" barplot or a barplot that shows proportions instead of counts.
(ii) Park determines position on the x-axis. Proportions within each bar are defined by tree Condition, which also determines the fill color. The graph is faceted by whether trees are native or non-native.



Ggplot code + deconstruction 2

Part A



Identify (i) what value(s) of `geom_---()` are involved in this graph and (ii) what you expect to be inside of `aes()`.

(i) This uses `geom_histogram()`. Not `geom_barplot`, because the x axis is quantitative!
(ii) `aes(x = body_mass, fill = species)`

Part B

(i) What type of graph will the following code produce, and (ii) how will all variables be mapped to aesthetic features? It might help to sketch an example of what the graph could look like.

```
penguins %>%
  count(year, island, species) %>%
  ggplot(aes(x = year, y = n, color = species)) +
  geom_line() +
  geom_point() +
  facet_wrap(~island)
```

(i) It will produce a line graph with points marked for each year (lines connect the points).
(ii) Position on the x axis is determined by `year`, position on the y axis is determined by `n`, and color of the points and lines is determined by `species`. We're also faceting by `island`, so a different plot will be produced for each island.